Introduction to Computational Quantum Chemistry: Modeling electronic properties

Part 2

Source: lectures by Dr. L Slipchenko

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Last time…

Hartree-Fock solutions:
- ignore electron correlations
- each electron interacts with the average field of other electrons

Computer solutions:
- Use K basis functions to represent orbitals, must be easy to integrate $\chi_i = \sum_{\mu=1}^{K} C_{\mu i}\varphi_\mu$ basis φ_μ, μ=1…K
- Solve for Hartree-Fock eigenvalues: $\underbrace{\hat{f}}_{\text{Fock operator}}(x_1)\chi_i(x_1) = \varepsilon_i \chi_i(x_1)$

$$\hat{f}\sum_\nu C_{\nu i}\varphi_\nu = \varepsilon_i \sum_\nu C_{\nu i}\varphi_\nu$$

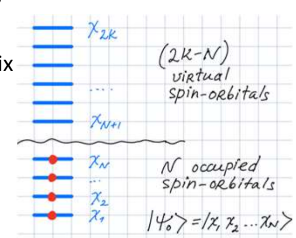
$$\sum_\nu C_{\nu i}\underbrace{\int dx_1 \varphi^*_\mu(x_1)\hat{f}(x_1)\varphi_\nu(x_1)}_{F_{\mu\nu}\text{ Fock matrix}} = \varepsilon_i \sum_\nu C_{\nu i}\underbrace{\int dx_1 \varphi^*_\mu(x_1)\varphi_\nu(x_1)}_{S_{\mu\nu}\text{ overlap matrix}}$$

$$\sum_\nu F_{\mu\nu}C_{\nu i} = \varepsilon_i \sum_\nu S_{\mu\nu}C_{\nu i}$$ Roothaan Equations

$$\mathbf{FC} = \mathbf{SC\varepsilon}$$

2K orthonormal spin-orbitals

2K equations: $\hat{f}_i|\chi_i\rangle = \varepsilon_i|\chi_i\rangle$



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The exact solution to N-electron problem

Given: N electrons and 2K spin orbitals.

The best solution:to combine all possible valid determinants $\Psi = \sum_i c_i \Phi_i$
and use the variational principle to minimize the ground state energy.
This model is called Full Configuration Interaction (FCI)
FCI is the exact solution of electronic SE in a given spin-orbital basis (with given K)

How big is a FCI wave function? The number of possible SDs is given as the number of times we can draw N spin-orbitals from a pool of 2K:

$$\#configurations = C_{2K}^{N} = \frac{(2K)!}{N!\,(2K-N)!} \approx 10^{N}$$

$\# config = C_{2K}^{N} = \frac{(2K)!}{N!(2K-N)!}$

N	$2K$	C_{2K}^{N}
2	10	45
4	20	4845
6	30	593775
8	40	$7.7 \cdot 10^{7}$
10	50	$1.0 \cdot 10^{10}$
20	100	$5.4 \cdot 10^{20}$

Today largest computations can deal with ~10^9 configurations

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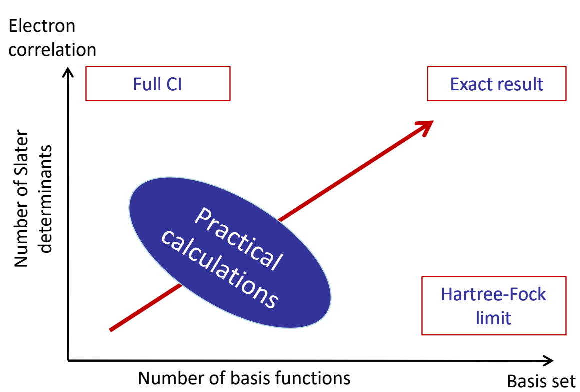
Theoretical model chemistries (John Pople)



Two approximations:

- We do not solve the exact equations → approximate treatment of electron-electron interactions (aka electron correlation)
- We do not achieve completeness in the electron expansion space (we do not solve inexact equations exactly) → basis set truncation error

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Electron correlation

Electron correlation can be recovered in post-Hartree-Fock calculations employing:

- Variational principle: Configuration Interaction methods (CISD, QCISD(T), etc)
- Perturbation theory: Moller-Plesset perturbation theory (MP2, MP4, etc.)
- Coupled-cluster ansatz (CCSD, CCSD(T), CCSDT, etc.)
- Density Matrix Renormalization Group (DMRG) theory
- Quantum Monte-Carlo (QMC)

or using the Kohn-Sham Density Functional Theory (DFT)

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Scaling and computational cost

accuracy

scaling

scaling	method	# of atoms
$N^{3(4)}$	Hartree-Fock, DFT	~500
N^5	second-order perturbation theory (MP2)	~100
N^6	configuration interaction singles and doubles (CISD); coupled cluster singles and doubles (CCSD)	~30
N^7	CCSD(T) and triples — "golden standard" of 1 kcal/mol = 43meV	~30
N^8	CISDT, CCSDT	~10
	……………	
$N!$	Full configuration interaction: FCI	

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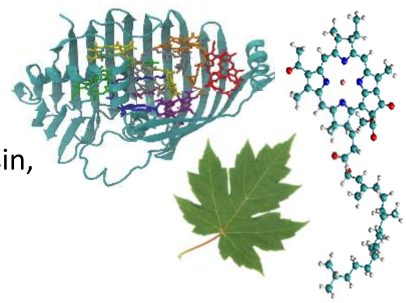
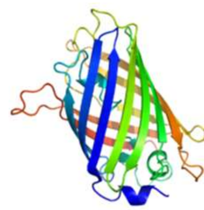
Electronic excited states

Natural and artificial photo-induced processes:

- Photosynthesis
- Photoactive proteins (GFP, rhodopsin, etc)
- Applications in photovoltaics
- Photo-induced catalysis
- Electronic spectroscopy
- …





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How to model excited states?

Several computational techniques:

- Configuration interactions (CI): CIS, CIS(D), CISD
  (Configuration Interaction Singles, CIS+ doubly excited corrections, CIS+Doubles)
- Time-dependent density functional theory (TD-DFT)
- Multi-configurational methods: MCSCF, CASSCF, CASPT2, MCQDPT, MR-CISD, etc.

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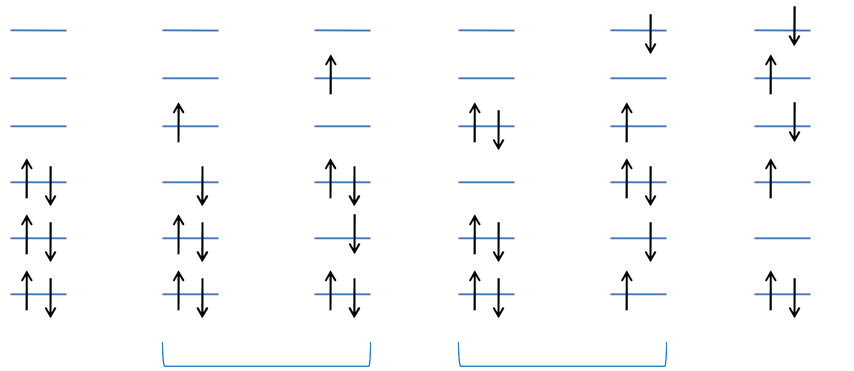
Excited Slater determinants



Φ_0 HF reference | Φ_S singly-excited | Φ_D doubly-excited | Φ_T triply-excited

$$\left|\Psi\right\rangle = a_0\left|\Phi_0\right\rangle + \sum_S a_S\left|\Phi_S\right\rangle + \sum_D a_D\left|\Phi_D\right\rangle + \sum_T a_T\left|\Phi_T\right\rangle + \ldots$$

exact wavefunction

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Configuration Interaction (CI)

Express solution as a linear combination of Slater determinants:

$$\Psi_{CI} = a_0\Phi_{HF} + \sum_S a_S\Phi_S + \sum_D a_D\Phi_D + \sum_T a_T\Phi_T + \ldots$$

Or: $\Psi_{CI} = \sum_{i=0} a_i\Phi_i$

Solve SE: $\hat{H}\Psi_{CI} = E\Psi_{CI}$

In Hilbert space: $\mathbf{\Psi_{CI}} = \begin{pmatrix} a_0 \\ \ldots \\ a_L \end{pmatrix}$ $\quad \mathbf{H} = \left(H_{ij}\right)$ $\quad H_{ij} = \left\langle\Phi_i\middle|\hat{H}\middle|\Phi_j\right\rangle$

$\mathbf{H\Psi_{CI}} = E\mathbf{\Psi_{CI}}$ Matrix equation

Finding energies of CI states E_i and wave function vectors a_{ij}

is a standard eigenvalue - eigenvector problem (or matrix diagonalization)

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How many determinants are in CI?

N electrons, K orbitals, n-tuple excitation $\rightarrow$ number of n-tuply excited determinants: $\binom{2N}{n}\binom{2K-N}{n}$

Number of singlet symmetry-adapted configurations (SAC) for H_2O in 6-31G*

Excitation level	# of excited SAC	Total # of SAC
1	71	71
2	2485	2556
3	40040	42596
4	348530	391126
5	1723540	2114666
6	5033210	7147876
7	8688680	15836556

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Truncated configuration interaction (CI)

CIS
K^5

CISD
K^6

CISDT
K^8

Full CI
~K!

$$\begin{array}{c} \\ \begin{pmatrix} \Phi_{HF} \\ \Phi_S \\ \Phi_D \\ \Phi_T \\ \Phi_Q \\ \vdots \end{pmatrix} \end{array} \begin{array}{c} (\Phi_{HF} \quad \Phi_S \quad \Phi_D \quad \Phi_T \quad \Phi_Q \quad \cdots) \\ \begin{array}{cccccc} E_{HF} & 0 & x & 0 & 0 & 0 \\ 0 & E_S & x & x & 0 & 0 \\ x & x & E_D & x & x & 0 \\ 0 & x & x & E_T & x & x \\ 0 & 0 & x & x & E_Q & x \\ 0 & 0 & 0 & x & x & \ddots \end{array} \end{array}$$

However, truncated CI (except for CIS) is not size consistent
i.e., E(A) + E(B) $\neq$ E(A+B) for non-interacting A and B

Solution: perturbation theory and coupled-cluster methods (CCSD(T), ADC, CC2, EOM-CCSD etc)

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How to diagonalize the matrix?

We need to extract one or a few of the lowest eigenvalues and eigenvectors of the H matrix →
iterative diagonalization techniques

Jacobi diagonalization (need to store the whole matrix)
Exact solution
Python: numpy.linalg.eig(), eigh() for Hermitian, real symmentric

Davidson diagonalization procedure (huge sparse matrices):
finds the lowest solutions to the H matrix by approximating them with solutions of a smaller matrix

In Python: scipy.linalg.eigsh()

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or using Kohn-Sham Density Functional Theory (DFT)

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Density Functional Theory (DFT)

Main object: electron density n(r)

Density of 1 electron: $n(r) = |\Psi(r)|^2$

N electrons: $n(r) = N\int \left|\Psi(r_1 = r, r_2, r_3, \ldots, r_N)\right|^2 dr_2 dr_3 \cdots dr_N$

Density depends on 3 coordinates while the wave function depends on 4N coordinates

The goal of DFT is to find a functional connecting electronic density and energy E = E[n(r)]

Hohenberg-Kohn theorem (1964):
The ground-state energy is completely determined by the electronic density:

$n(r) \dashrightarrow E_{gs}$

(but the connection is unknown)

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Orbital-free DFT

Electron-nuclear attraction: $E_{ne}[n(r)] = -\Sigma_A^{nuclei} \int \frac{Z_A}{|r - R_A|} n(r) dr$

Electron-electron interaction: Coulomb $J[n(r)] = \frac{1}{2}\int \frac{n(r)n(r')}{|r - r'|} dr dr'$

Consider a uniform electron gas (UEG)

Kinetic energy (Thomas & Fermi, 1927): $T_{TF}[n(r)] = \boxed{c_F} \int n^{\frac{5}{3}}(r) dr$

Exchange energy (Dirac, 1929): $K_D[n(r)] = -\boxed{c_X} \int n^{\frac{4}{3}}(r) dr$

Analytical constants

Thomas-**F**ermi-**D**irac theory:

$$E_{TFD}[n] = T_{TF}[n] + E_{ne}[n] + J[n] + K_D[n]$$

Exact for UEG, but does not predict bonding in molecules!

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Orbital-free DFT

Why Thomas-Fermi-Dirac theory does not work?

$T_{TF}[n]$ underestimates kinetic energy by ~10%

Ne atom: E_{kin} = 128.9 a.u. = 27 eV
$E_{exchange}$ = - 12.1 a.u. = -2.53 eV
E_{corr} = - 0.4 a.u. = 0.083 eV

kT_{room} = 0.025 eV

Kinetic density functional is the biggest problem!

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Kohn-Sham DFT

Solution: Kohn & Sham approach

Represent kinetic energy as two terms:
one will be exact through orbitals (Slater Determinant term, T_{SD}),
second – small correction

Price:

- back to orbitals (from 3 to 4N coordinates)
- correlation functional reappears as a separate term E_{xc}

Kohn-Sham functional:

$$E_{KS}[n] = \underbrace{T_{SD}[n]}_{\text{slater determinant kinetic energy}} + \underbrace{E_{ne}[n]}_{\text{Electron-nuclear attraction}} + \underbrace{J[n]}_{\text{(Density) Electron-electron attraction}} + E_{xc}[n]$$

where E_{xc} is defined as:

$$E_{xc}[n] = \underbrace{(\overbrace{T[n]}^{\text{Total kinetic energy}} - T_{SD}[n])}_{\text{kinetic correlation energy}} + \underbrace{(\overbrace{E_{ee}[n]}^{\text{Total e-e attraction (unknown)}} - J[n])}_{\text{potential correlation \& exchange energies}}$$

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Kohn-Sham DFT

$$E_{KS}[n] = T_{SD}[n] + E_{ne}[n] + J[n] + \boxed{E_{xc}[n]}$$

compare with Hartree-Fock energy:

$$E_{HF}[\{\chi\}] = T + E_{ne} + J \boxed{- K}$$

Using variational principle, Kohn-Sham functional results in a self-consistent problem $\hat{f}^{KS}\chi_i = \varepsilon_i \chi_i$

with $\hat{f}^{KS}(x_1) = \hat{h}(x_1) + \Sigma_j \hat{J}_j(x_1) + V_{xc}(x_1)$

exchange-correlation potential $V_{xc} = \dfrac{\delta E_{xc}}{\delta n}$

Kohn-Sham theory is exact if V_{xc} is known

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Electron self-interaction energy

Hartree-Fock electron-electron interaction: $J - K =$

$$\int dx_1 dx_2 \chi_i^*(x_1)\chi_j^*(x_2)\frac{1}{r_{12}}\chi_i(x_1)\chi_j(x_2)$$
$$-\int dx_1 dx_2 \chi_i^*(x_1)\chi_j^*(x_2)\frac{1}{r_{12}}\chi_i(x_2)\chi_j(x_1)$$

if $x_1 = x_2$ → $J - K = 0$

Electron self-interaction energy is zero!

DFT: electron-electron interaction is given by $J[n] + E_{xc}[n]$

$$J[n(r)] = \frac{1}{2}\int \frac{n(r)n(r')}{|r - r'|} dr dr' \neq 0$$ for 1 electron

$E_{xc}[n]$ = ?

It is hard to cancel self-interaction with non-HF exchange functional

Consequences:
underestimation of reaction barriers, energies of charge-transfer states, band gaps, dissociation energies of ions, Rydberg states

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Functionals

Perdew-Jacob's ladder of approximations: $n \rightarrow \nabla n \rightarrow \nabla^2 n \rightarrow \ldots$

1. Local density approximation (**LDA**): functionals depend on density only
   - exchange and correlation functionals from uniform electron gas (UEG)
   - LDA is exact for UEG!
   - no fitted parameters
   - ~10% error in exchange energy
   - accuracy similar to Hartree-Fock

UEG: uniform electron gas

2. Gradient-corrected methods: Generalized Gradient Approximation (GGA)
   functionals depend on density and density derivatives

e.g. Becke exchange functional:

reduces error in exchange by 2 orders!

$$\varepsilon_x^{B88} = \varepsilon_x^{LDA} + \Delta\varepsilon_x^{B88}$$

$$\Delta\varepsilon_x^{B88} = -\beta n^{\frac{1}{3}} \frac{x^2}{1 + 6\beta x sinh^{-1} x}$$

fitted to rare gases

$$x = \frac{|\nabla n|}{n^{\frac{4}{3}}}$$

BLYP, PW86, PW91, PBE

See details: https://manual.q-chem.com/4.3/sect-DFT.html

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Functionals

3. Meta-GGA
   - include Laplacian $\nabla^2 n$ or orbital kinetic energy density
   - BR, B95, HCTH, TPSS, …
   - Not a significant improvement in accuracy

4. Back to self-interaction problem: hybrid or hyper-GGA functionals
   include a portion of the Hartree-Fock exchange

$$E_{xc}^{B3LYP} = (1-a)E_x^{LDA} + aE_x^{HF} + b\Delta E_x^{B88} + (1-c)E_c^{LDA} + cE_c^{LYP}$$

B3LYP: a=0.20, b=0.72, c=0.81
PBE0
HF & LDA exchange produce error cancelation

5. Double hybrid methods:
   - add MP2 (perturbation theory) correlation
   - high computational cost

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DFT: summary

Pros

- Computational cost & scaling: similar to Hartree-Fock
- Superb geometries and frequencies
- Good energetics for covalent systems
- Smaller spin-contamination in open-shell systems

Cons

- Every functional goes nuts on some system or other
- Missing long-range correlation (van der Waals interactions) → check out functionals with D (dispersion) corrections
- DFT should not be applied to metals without careful benchmarking!
- Errors due to electron self-interaction
- States of different multiplicities (HF favors high spin, DFT favors low spin)

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CIS vs TDDFT

	CIS	TDDFT (time-dependent DFT)	
accuracy:	~1 eV error	0.1 - ? eV	(functional dependent)
computational scaling:	N^5	N^5	(bigger prefactor)

model single excitations only

TDDFT: Potential trouble with charge-transfer states & Rydberg states → use long-range corrected & hybrid functionals

different multiplets

CIS: high-spin states too low in energy

TDDFT: high-spin states too high in energy → use long-range corrected & hybrid functionals

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